ZCAS University

CCS2301 - OPERATING SYSTEM CONCEPTS

END OF SEMESTER EXAMINATION

1ST JUNE 2023

16:30 HRS - 19:30HRS

TIME ALLOWED: **WRITING – THREE HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

As a basic definition, we may state that a deadlock is a situation in which two computer processes sharing the same resource are effectively preventing each other from accessing the resource, resulting in both programs ceasing to function.

Here is a simple illustration:

Process 1 requests resource A and receives it.

Process 2 requests resource B and receives it.

Process 1 requests resource B and is queued up, pending the release of B.

Process 2 requests resource A and is queued up, pending the release of A.

Learning how to deal with deadlocks has a major impact on the development of operating systems and the management of processes / resources.

i) A **single-user operating system** is a type of operating system developed and intended for use on a computer system that will only have a single user at any given time. There are two (2) types of a single-user operating system:

   a) Single-User Single-Tasking operating system, and

   b) Single-User Multi-Tasking operating system.

   Discuss the challenges associated with a deadlock in each of the two (2) types of a single-user operating systems stated above. **(8 Marks)**

ii) Identify and briefly explain each of the four (4) common conditions which if present, at the same time, in a computer system could lead to a **deadlock situation**. **(12 Marks)**

iii) Under the concept of "**Deadlock Prevention**", it is stated that a violation of any of the four (4) conditions necessary conditions identified in (ii) above could prevent a deadlock. Discuss the implementation of each of these four (4) violations and briefly explain the challenges (if any) that are associated with each implementation. **(12 Marks)**

iv) "**Deadlock Avoidance**" can be done with the use of the Banker's Algorithm. Explain the Banker's Algorithm and illustrate how it could be used in a "Deadlock Avoidance" strategy. **(8 Marks)**

(Total: 40 Marks)

SECTION B: **Attempt any THREE questions in this section.**

Question 2

i) What is an operating system? (2 Marks)

ii) Identify, state and discuss any FOUR (4) of an Operating System's major roles in a computer system. (8 Marks)

iii) Explain the difference between a "Program" and a "Process". (4 Marks)

iv) Explain the difference between CPU-Bound Processes and I/O-Bound Processes. (4 Marks)

v) What is meant by the term "Volatile Device". Give an example. (2 Marks)

(Total: 20 Marks)

Question 3

i) It is often said that increasing the size of Random Access Memory (RAM) in a computer system would normally result into an increased performance in terms of processing speeds. Explain how this increased performance arising from an increase in the size of RAM is achieved. (4 Marks)

ii) During its execution, a process goes through multiple states. Identify and discuss any four (4) of these states. (8 Marks)

iii) A Swapping Process is a mechanism that temporarily moves some processes out of the main memory to the secondary storage and back to the main memory. Explain the circumstances which may necessitate the need for this swapping process in a computer system. (4 Marks)

iv) Let us assume that a standard hard disk where swapping takes place has a data transfer rate of about 1 MB per second. How much time in seconds would a 4096KB user process take to be swapped from the main memory to this standard hard disk and back to main memory? [You should clearly show your working] (4 Marks)

(Total: 20 Marks)

Question 4

There are various algorithms which are used by the Operating System to schedule the processes on the processor in an efficient way. A Process Scheduler schedules different processes to be assigned to the CPU based on particular scheduling algorithms.

i) Apart from the Shortest Remaining Time First (SRTF) algorithm, identify and discuss any other three (3) of the commonly used process scheduling algorithms. (9 Marks)

ii) The table below shows the Process ID, the Arrival Time and the Burst Time for each Process. Using the SRTF, you are required to:

a) determine the Completion Time, Turn Around Time and Waiting Time for each Process. (9 Marks)

b) calculate the Average waiting time of the system. (2 Marks)

Process ID	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
1	0	9			
2	1	4			
3	2	9			

(Total: 20 Marks)

Question 5

i) Memory Fragmentation can be divided into two (2) categories: Internal Fragmentation and External Fragmentation. Discuss each of the two types and state how each one of them may be reduced. (6 Marks)

ii) Paging and Segmentation are two different techniques that are used for memory management. Discuss the two memory management techniques, clearly stating the differences between them. (6 Marks)

iii) Explain the concept of Demand Paging. (2 Marks)

iv) Clearly state and discuss **one advantage** and **one disadvantage** associated with the Demand Paging Concept. (6 Marks)

(Total: 20 Marks)

.

END OF EXAMINATION